

Python Programmation

CM4 : First Course Recap

Julien Bezançon

julien.bezancon@univ-lorraine.fr / <https://julienbez.github.io>



During this recap, we will correct exercises
from previous practical sessions.

Short exercises: using the board.

Long exercises: by group (10 groups max).

Python Basics

- Variables and types
- Operators
- Console input/output
- File manipulation

Control Flow

- Statements
- Loops



Data structures

- Lists
- Dictionaries
- Tuples
- Sets

Advanced Python Stuffs

- Advanced loops
- Functions



Functions

- Function definition and calling
- Parameters and arguments
- Variadic arguments
- Python scopes

Advanced strings

- Special characters handling
- String formatting



Before we start, here are some common mistakes
I saw in your notebooks.

Respect the Rules

- Here is the intended name of a notebook: TD1_LASTNAME_LASTNAME.ipynb
- A TD without your name $\Rightarrow$ points denied
- A TD without the name of your binom $\Rightarrow$ points denied

Do What I Ask for

- "Write a function to..." $\Rightarrow$ function expected

```
a = 1
```

```
b = 2
```

```
my_add_function = ( a + b)
```

Indentation Error

- Stick to one indentation scheme (2 spaces or 4 spaces for instance, not both)

```
def my_add_function(a,b):  
    return a + b
```

```
def my_add_function(a,b):  
    return a + b
```

Indentation Error II, the Return

- "return" should not be in a loop, unless you add a condition

```
def find_mystery_number():  
    guess = 1  
    while hot_or_cold(guess) != "bingo bango!":  
        guess += 1  
    return guess
```

Variables Names Matters

- Don't use variable names which are used in Python builds-in

```
list = []  
list.append(i)  
len(list)  
list[i]
```

- Be careful with function calling, names must be respected

```
def convert_fahr_to_cels(def_fahr):  
    # do stuff...
```

```
fahrenheit_to_celsius(fahrenheit)
```

"Inspiration" VS plagiarism

- DO NOT COPY/PASTE SOMEONE ELSE CODE !!!

Influenced by "some url"

```
def collatz_number(n):  
    # do stuff...
```

Other Remarks

- Always make sure that you call your functions or print the result in a cell
- If I have to edit your notebook to see a cell output, I will subtract some points
- Be careful when sending your TDs, I got two empty TDs...

Now, let's proceed to the corrections !

Infinite Loop

```
def find_mystery_number():
    guess = 10
    while True:
        result = hot_or_cold(guess)
        if result == "bingo bango!":
            return guess
        if result == "you're hot!":
            break
        guess += 9
    guess -= 4
    while True:
        result = hot_or_cold(guess)
        if result == "bingo bango!":
            return guess
        guess += 1
```

String Warning

write your code here

```
dict_contents=({"chimps": 42, "peanuts": 0})
```

```
def silly_messenger(**d):
```

```
    print(f"As for chimps, I've got {d['chimps']}, sadly.")
```

```
silly_messenger(**dict_contents)
```
